DELHI PUBLIC SCHOOL JANKIPURAM
ANNUAL EXAMINATION: (2024-2025)
SUBJECT: MATHEMATICS
GRADE: IX

SET–I

Time – 3 hrs. M.M. 80

Name- Mohd. Hifzaan Khan Roll No- 20 Date- 19 Feb'25

General Instructions:

Read the following instructions carefully and follow them:

1. *This question paper contains 38 questions.*
2. *This Question Paper is divided into 5 Sections A, B, C, D and E.*
3. *In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20are Assertion- Reason based questions of 1 mark each.*
4. *In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.*
5. *In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.*
6. *In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.*
7. *In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.*
8. *All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.*
9. *Draw neat and clean figures wherever required.*
10. *Take $\pi = 22/7$ wherever required if not stated.*
11. *Use of calculators is not allowed.*

Section -A

This section comprises multiple choice questions (MCQs) of 1 mark each.

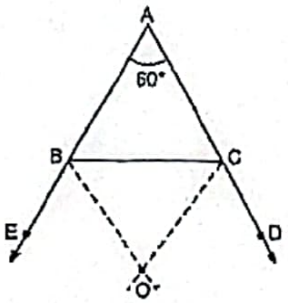
1. Which of the following rational numbers is equivalent to decimal that terminates.
   (a) $\frac{1}{3}$ (b) $\frac{8}{9}$ (c) $\frac{3}{8}$ (d) $\frac{5}{6}$

2. If we multiply or divide both sides of a linear equation with a non – zero number, then the solution of the linear equation
   (a) changes (b) remains the same
   (c) changes in case of multiplication only (d) changes in case of division only

3. Abscissa of all the points on the X–axis is
   (a) 0 (b) 1 (c) 2 (d) any number

4. Euclid's first postulate is
   (a) all right angles are equal to one another
   (b) a circle may be described with any centre and any radius.
   (c) a terminated line can be produced indefinitely.
   (d) a straight line may be drawn from any point to any other point

5. In figure BO & CO are the bisectors of external angles ∠ CBE and ∠ BCD respectively of ΔABC. Then ∠ BOC is equal to



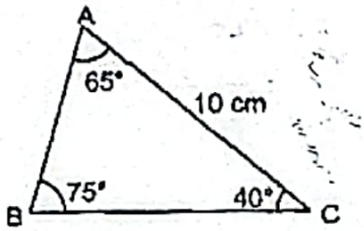
(a) 120° (b)30° (c) 60° (d) 80°

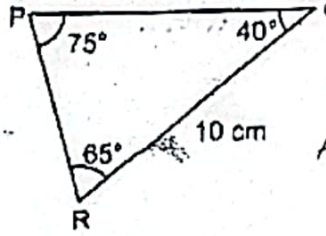
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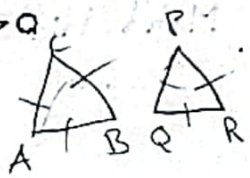
6. Meena drew a figure and named it $\overrightarrow{AB}$. Which of the following best describes the figure Meena drew?
(a) It is an angle (b) It is a line segment
(c) It could be a line segment or a ray (d) It is a ray

7. If AB = QR, BC = RP and CA = PQ then
(a) ΔABC ≅ ΔPQR (b) ΔCBA ≅ ΔPRQ (c) ΔBAC ≅ ΔRPQ (d) ΔBCA ≅ ΔPQR.

8. Consider the triangles shown.





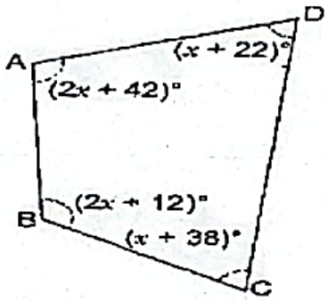


Which congruency criteria can be used to show that triangle ABC and RPQ are congruent.
(a) SAS (b) SSS (c) ASA (d) AAA

9. In a parallelogram ABCD, if ∠ A = 75° then ∠ C is equal to
(a) 15° (b) 105 (c) 75° (d) none of these

10. Consecutive angles of a parallelogram are
(a) Complementary (b) supplementary (c) Equal (d) none of these

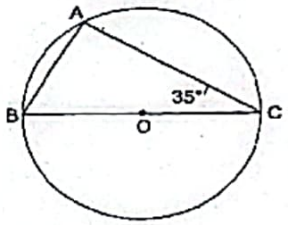
11. A quadrilateral ABCD is shown below.



$2x + 2x + x + x = 6x$
$42 + 22 + 12 + 38 = 114$
$6x = 180 - 114$
$x = \frac{66}{6}$
$x = 11$

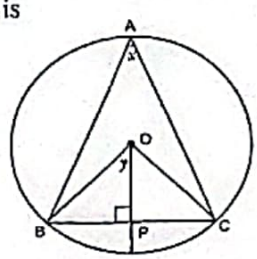
What is the measure of ∠ BAD?
(a) 64° (b) 84° (c) 104° (d) 124°

12. In the given Figure, O is the centre of the circle. If ∠ACB = 35°, then ∠ABC is equal to



(a) 35° (b) 55° (c) 15° (d) none of these

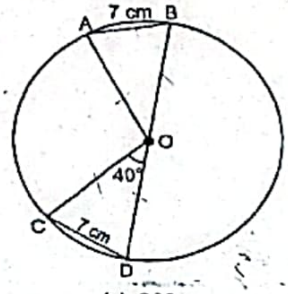
13. In the given figure, O is the centre of a circle if ∠BAC = x, OP bisects ∠BOC and ∠BOP = y, then the relation between x and y is



(a) x=2y (b) x=y (c) $x = \frac{y}{2}$ (d) none of these

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14. In the given Figure, A circle with centre O is given, Find the measure of ∠ AOB ?



(a) 30° (b) 40° (c) 80° (d) 140°

15. The sides of a triangle are 11 m, 60 m, 61 m . The altitude to the smallest side is
(a) 11 m (b) 50 m (c) 60 m (d) 66 m

16. The length of the hypotenuse of an isosceles triangle with area 72 sq cm is
(a) 12 cm (b) $12\sqrt{2}$ cm (c) 24 cm (d) 12.5 cm

17. The height of a cone is 16 cm and base radius is 12 cm. Its slant height is
(a) 10 cm (b) 15 cm (c) 20 cm (d) 8 cm

18. The range of the data :
25 , 18 ,20 , 22 , 16 , 6 , 17, 15 , 12 , 30 , 32, 10, 19 ,8 ,11 ,20
(a) 10 (b) 15 (c) 18 (d) 26

Direction: In the question number 19 and 20,
a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option:

(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)
(b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)
(c) Assertion (A) is true but reason (R) is false.
(d) Assertion (A) is false but reason (R) is true.

19. **Assertion (A)** : In ΔABC, ∠A=43°and ∠C=70° then the measure of ∠B is 67°
Reason(R) : Sum of angle of triangle is 180°

20. **Assertion(A)** : The class mark of the class interval 90-120 is 105
Reason(R) : Class mark = (upper limit + lower limit) / 2

Section –B

This section comprises very short answer (VSA) type questions of 2 marks each.

21. a) Simplify : $(3\sqrt{5} - 5\sqrt{2})(4\sqrt{5} + 3\sqrt{2})$

OR

b) Which is greater $\sqrt[3]{3}$ or $\sqrt[4]{4}$?

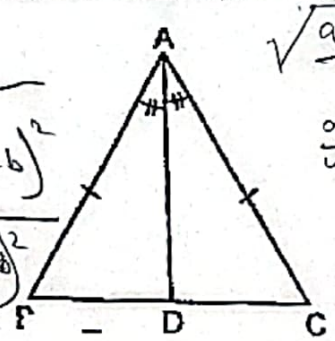
22. ABCD is a parallelogram. If ∠ DAB= 60°and ∠ DBC = 80°, find ∠ CDB.

OR

In a rectangle ABCD, ∠ BAC = 32°, find the measure of ∠ DBC.

23. If the ratio between two complementary angles is 5:4, find the angles.

24. In figure, if AB = AC, BD = DC and AD bisects ∠A , then find ∠ADB.



25. The perimeter of an isosceles triangle is 32 cm. The ratio of the equal sides to its base is 3:2. Find the area of the triangle.

Section –C

This section comprises short answer (SA) type questions of 3 marks each.

26. a) If $a = 2 + \sqrt{3}$, find the value of $a - \frac{1}{a}$.

OR

b) Find the value of x, if $\left(\frac{6}{5}\right)^{x}\left(\frac{5}{6}\right)^{2x} = \frac{125}{216}$.

27. Without finding the cubes, factorise: $(x-2y)^3 + (2y-3z)^3 + (3z-x)^3$

28. Using factor theorem, factorise the polynomial $x^3 + x^2 - 4x - 4$

29. Give the equation of two lines passing through (-2,-4).How many more such lines are there, and why?

30. a) Find the volume of a sphere whose surface area is 154 cm^2.

OR

b) A conical tent is 10 m high and the radius of its base is 24 m .Find
   (i) slant height of the tent
   (ii) cost of the canvas required to make the tent , if the cost of 1m^2 canvas is Rs 70.

31. Find the coordinates of the points
   (a) Which lies on both $X-$axis and $Y-$axis?
   (b) Whose abscissa is 5 and lies on x-axis?
   (c) Whose ordinate is -4 and lies on y-axis?

Section –D

This section comprises long answer (LA) type questions of 5 marks each.

32. a) If $x - y = 5$ and $xy = 84$, find the value of $x^3 - y^3$.

OR

b) If $z^2 + \frac{1}{z^2} = 14$, find the value of $z^3 + \frac{1}{z^3}$.

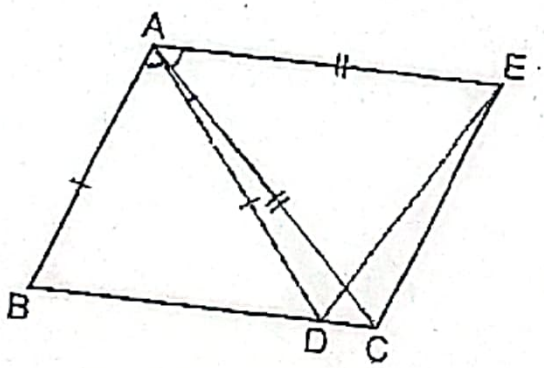
33. Draw the graph of the equation $3x + 4y = 12$ and find the coordinates of the points of intersection of the equation with co-ordinate axes.

34. The diameter of a sphere is decreased by 25%. By what percent does its curved surface area decreased?

35. a) Prove that if in two triangles , two angles and the included side of one triangle are equal to two angles and included side of other triangle, then two triangles are congruent.

OR

b) In figure , AC = AE ,AB = AD and $\angle BAD = \angle EAC$. Show that BC = DE.



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Section- E

This section comprises 3 case study-based questions of 4 marks each.

36. **Case-Study – 1** Manik and Dhruv are bench - mates in the class. In the mathematics class, Manik was finding that it was difficult to simplify $\frac{1}{(\sqrt{5}-\sqrt{2})}$. His bench-mate Dhruv gave him a clue to rationalize the denominator by taking the conjugate of $(\sqrt{5}-\sqrt{2})$. Manik simplified the expression and thanked Dhruv for the help. Dhruv also gave him the approximate value of $\sqrt{5}=2.236$ and $\sqrt{2}=1.414$ to find the approximate value of the expression.



Based on the above information answer the following questions:

36.1 What is the conjugate of $(\sqrt{5}-\sqrt{2})$? (1)

36.2 Where should we multiply the conjugate of $\sqrt{5}-\sqrt{2}$ to rationalize? (1)

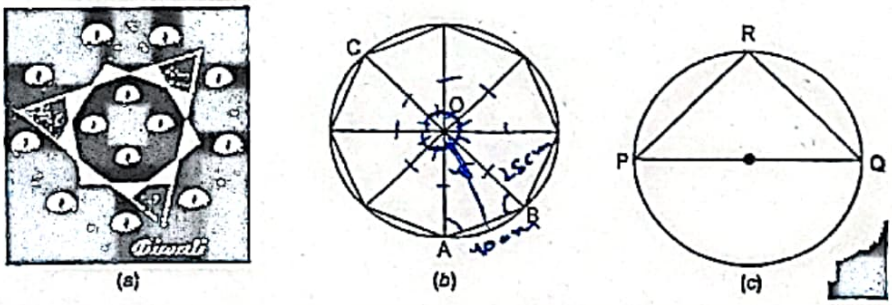
36.3 What is the simplified form of the expression that Manik found out? (2)

OR

What is the approximate value of the expression did Manik find after putting the values $\sqrt{2}=1.414$ and $\sqrt{5}=2.236$

37. **Case-Study – II**

Rangoli hold a significant role in everyday life of a Hindu household .It has different names based on the state and culture. It represents happiness, positivity and liveliness of a household. Kajal made two rangoli in front of her house. She started one with a circle of diameter 50cm and made triangles as shown in the figure .In another circle, she drew one isosceles triangle as shown in the figure.



(a) (b) (c)

On the basis of the above information, solve the following questions:

37.1 If Kajal divides the circle into eight equal parts, find the angle subtended by each arc at the centre. (1)

37.2 In figure, the length of the chord AB of the circle is 40 cm. Find its distance from centre. (1)

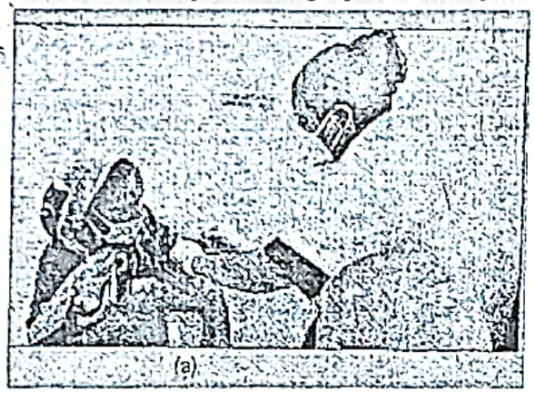
37.3 In figure, If PQ is the diameter, then find the measure of ∠PRQ. 90° (2)

OR

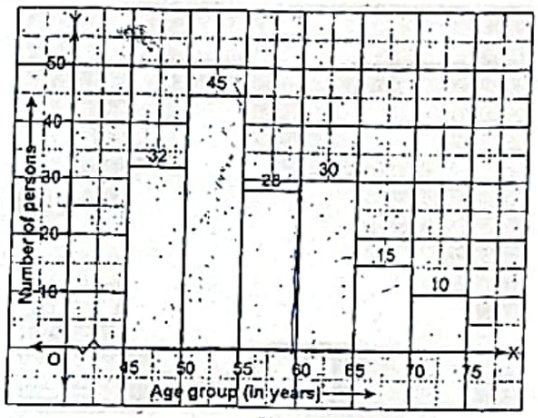
In figure, If PQR is an isosceles triangle, then find the measure of ∠RPQ.

38. **Case-Study – III**

The histogram here shows the number of persons in various age groups who were vaccinated in a private hospital during a particular day.



(a)



(b)

On the basis of the above information, solve the following questions:

38.1 In which age group were the maximum number of persons vaccinated. 50-55 (1)

38.2 How many persons above 60 years of age were given vaccination? 55 (1)

38.3 What is the class mark of the first class? 47.5 (2)

OR

38.3 How many more persons were vaccinated in the age group 50 to 55 years as compare to 45 years to 50 years?

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